

# **GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE**

## **I. INTRODUCTION**

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as “the 2021 Recommendation”) that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO’s Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

## **II. WHAT ARE THE AIMS OF THIS CONSULTATION?**

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

## **III. HOW TO FILL IN THE QUESTIONNAIRE?**

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and

innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at <https://www.unesco.org/en/open-science/toolkit> and <https://unesdoc.unesco.org/ark:/48223/pf0000387324>.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

- (i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.
- (ii) Email: the questionnaire can be completed electronically and sent to: [openscience@unesco.org](mailto:openscience@unesco.org)

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

#### IV. DRAFT QUESTIONNAIRE

##### A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country\*: [Romania](#)
- Organization(s) or entity(s) responsible for the preparation of the report\*:  
Name: [UEFISCDI](#)  
Website: <https://uefiscdi.gov.ro>  
Email address\*:  
Phone number\*:

Please describe the role/mandate of your organization:

The Executive Agency for Higher Education, Research, Development and Innovation Funding (UEFISCDI) is the main funder for competitive research in Romania, and a policy adviser for science, innovation and higher education policies, under the ultimate authority of the Ministry of Education and Research (MEC). As a funding agency, UEFISCDI supports exploratory and applied research, in all branches of science and the humanities, and it funds research projects on a competitive basis. The Agency's prerogatives in the research area pertain to the implementation of the majority of Programmes under the National Research, Development and Innovation Plan (PNCDI IV), which is the main

instrument for implementing the [National Strategy on Research, Innovation, and Smart Specialization for 2022–2027](#).

UEFISCDI, plays a central role in advancing Open Science policies and practices at the national level. The institution is active in the Open Science subject since 2014, taking part in key initiatives carried at European level.

In 2019, UEFISCDI created the [Open Science Knowledge Hub Romania](#) (OSKH Romania), as a national hub aimed at promoting and guiding Open Science across the country. The institution, though this dedicated team, has played a crucial part in coordinating the National Open Science Strategic Framework and providing policy recommendations which have been integrated into Romania's National Strategy on Research, Innovation, and Smart Specialisation (2022–2027) and further detailed into the [White Paper on the Transition to Open Science 2023-2030](#) (as part of the project *Increasing the capacity of the RDI system to respond to global challenges*"- funded through European Structural and Investment Funds). Within the same project, UEFISCDI developed and launched in December 2022 a dedicated national [Open Science portal](#) and created an Open Science Community in the [BrainMap](#) platform.

The institution is actively engaged in international collaborations, and it ensures Romania aligns with global Open Science efforts. The work carried out for Open Science is linked to UNESCO's initiatives also through a close collaboration with the [UNESCO Chair On Science and Innovation Policies – SNSPA](#).

- Officially designated contact person(s) who completed this survey:

Full name\*: [Alina Irimia](#)

Position\*: [Open science projects coordinator](#)

Email address\*:

- Full Name(s) of designated official(s) certifying the report:

[National Research Authority \(recently established by new legislation; official contact person responsible for Open Science not yet determined\)](#)

- Brief description of the consultation process established for the preparation of the report:

The report on the implementation of the 2021 Recommendation on Open Science was developed through extensive desk research, analysis of relevant policies, reviewing different consultation results and relevant projects to assess the current state of open science implementation. Consultations were conducted with various departments at UEFISCDI to gather institutional insights, alongside engagements with the Ministry of Research and Education to align national policies with the UNESCO framework. Additionally, discussions with the Romanian National Commission for UNESCO (CNR UNESCO) provided further perspective on international commitments and best practices, ensuring a robust and informed report. In the preparation of the report, guidance from UNESCO's resources, including the UNESCO Open Science Toolkit and the first edition of the UNESCO Open Science Outlook (published in December 2023), were also revised.

- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:

- Name of the organization\*: Ministry of Education and Research, The National Research Authority

Website\*: [www.edu.ro](#) , [www.mcid.gov.ro](#)

Email address\* not yet officially designated

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

Name of the organization\*: The Romanian National Commission for UNESCO

Website\*: <https://www.cnr-unesco.ro/>

Email address\*

Sector: ☒ Public ☐ Private ☐ Civil Society ☐ Other

## B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

### 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

☒ YES / NO

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

**Ministry of Education and Research, The National Research Authority**

**UEFISCDI**

**The Romanian National Commission for UNESCO**

**UNESCO Chair on Science and Innovation Policies**

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ YES / NO

If yes, which language(s)?

**Romanian**

1.3 Have there been awareness raising activities on open science, including all its key elements<sup>i</sup>, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) [16.f](#), [16.g](#), [16.h](#))

☒ YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

#### 1) Organized (starting 2020):

- (In person) **Rethinking Research Assessment Conference**  
**Organized by:** UEFISCDI, in collaboration with the Open and Universal Science project ([OPUS](#)), [Science Europe](#), the Coalition for Advancing Research Assessment ([CoARA](#)), and the [UNESCO Chair on Science and Innovation Policies \(SNSPA\)](#).

**Date:** November 14, 2024  
**Number of Attendees:** 101

**Link:** [Event Details](#)

- (Online) **Romania National Roadshow - Collaborative FAIR Data Practices & Tools**  
**Organized by:** UEFISCDI, as part of the FAIR-IMPACT project, through the Open Science Knowledge Hub (OSKH)  
**Date:** October 8, 2024, from 10:00 to 12:00 CEST  
**Link:** [Event Details](#)
- (Hybrid) **Open Education & Science Workshop 2024. Challenges in Open Science and Education**  
**Organized by:** Politehnica University of Timișoara, through the Department of ID/IFR and Digital Education, in collaboration with the E<sup>3</sup>UDRES<sup>2</sup> Alliance of European Universities and the Ent-r-e-novators project, with support from IEEE Romania.  
**Date:** Friday, March 8, 2024, part of Open Education Week 2024  
**Number of Attendees:** 360 (on-line), 80 (in person)  
**Link:** [Event Details](#)
- (Online) **Open Science – Updates on Open Access, Research Assessment, and FAIR Data**  
**Organized by:** Open Science Knowledge Hub (OSKH) – UEFISCDI and the UNESCO Chair on Science and Innovation Policies  
**Date:** October 13, 2023  
**Number of Participants:** Over 190  
**Link:** [Event Details](#)
- (in person) **Research Assessment and Open Science** – informal discussions with the Romanian policy makers:  
**Organized by:** the Romanian Academy  
**Date:** 28 April 2023  
**Number of Attendees:** 25 participants from the Romanian Academy as well as from different research organizations.
- (Hybrid) **Dissemination Event and Setting up the Romanian Initiative for Open Science Cloud (RO-NOSCI)**  
**Organized by:** UEFISCDI, [ICI-Bucharest](#), The National Institute for Research and Development for Physics and Nuclear Engineering "Horia Hulubei" – [IFIN-HH](#)  
**Date:** June 17, 2021  
**Number of Attendees:** around 60  
**Link:** [Event Details](#)
- (Online) **FAIRsFAIR Roadshow – Romania**  
**Organized by:** FAIRsFAIR, in collaboration with the Open Science Knowledge Hub Romania, UEFISCDI  
**Date:** July 7, 2021  
**Number of Attendees:** around 120  
**Link:** [Event Details](#)
- Webinar: **"OpenAIRE Romania NOAD Support for the Open Science and EOSC Transition"**  
**Organized by:** UEFISCDI as the OpenAIRE NOAD Romania  
**Date:** February 26, 2021  
**Link:** [Event Details](#)
- **EOSC and Open Science Dissemination** (Part of the NI4OS Project)  
**Organized by:** UEFISCDI & ICI-Bucharest  
**Date:** October 27, 2020  
**Number of Attendees:** Approximately 40  
**Links:** [NI4OS Event Overview](#), [UEFISCDI Event Page](#)
- Webinar: **"Open Science with a Focus on Research Data Management (RDM)"**  
**Organized by:** UEFISCDI as the RDA Node  
**Date:** October 6, 2020  
**Number of Attendees:** 240 online attendees  
**Link:** [Event Details](#)

- Webinar: "**The Importance of Open Science for Societal Challenges – OS & COVID-19**"  
Organized by UEFISCDI as the OpenAIRE NOAD and RDA Node Romania  
Date: April 14, 2020  
Number of Attendees: 50  
Link: [OpenAIRE Blog on the Event](#)
- Face-to-Face event: "**The Future of Open Science in Romania**", kick-off of the consultation process for a National Open Science Strategic Framework  
Organized by UEFISCDI  
Date: March 10, 2020  
Number of Attendees: around 40  
Link: [Event Details](#)

## 2) Foreseen:

- Dissemination and consultation with the RDI community on a DMP template (adapted based on the Science Europe recommendations) – April 2025 (part of the [Opus project](#));
- Dissemination regarding new practices (strategy implementation): piloting new requirements and evaluation criteria related to Open science in funding instruments (part of the [Opus project](#))
- Dissemination regarding new practices (strategy implementation) and consultation with regards to an Openness Profile to be embedded into a new researcher profile part of the BrainMap platform (part of the [GraspOS project](#)).

Apart of these, constant dissemination is carried out by the OSKH team at UEFISCDI through a specific section in the national OS portal (<https://www.open-science.ro/stiri-si-evenimente>), UEFISCDI dedicated webpage (<https://uefiscdi.gov.ro/noutati-open-science>) and through the Open Science Community part of the [BrainMap](#) Platform. Regular (2-3 times a week) updates on social media/ BrainMap OS Community/ OS Portal/ UEFISCDI website. Close connection to UNESCO – linking all work with the UNESCO Chair on Science and Innovation Policies.

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science<sup>ii</sup>, in publicly funded research? (*ref.:* [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

**YES** / NO

**If yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

- A pilot of new requirements and evaluation criteria related to Open science has started since 2023: Requirements related to Data Management Plans (DMPs) and Open Science have been included in the funding instrument "Centers of Excellence". Applicants have been required to describe how open science practices (e.g. Open Access to publications, research data management, citizen science, and other) are implemented in their project to enhance its objectives. If no open science practices are applied, a justification is required. Additionally, the applicants have been asked to describe the research data management (RDM) aligned with FAIR principles (Findable, Accessible, Interoperable, and Reusable). If funded, a Data Management Plan (DMP) is required within the first six months of the project's implementation. More info: <https://uefiscdi.gov.ro/centre-de-excelenta-coe-coe2023-747>

The requirements and criteria have been extended to other funding instruments which are planned to be launched in near future:

- Complex Frontier Research Projects: <https://uefiscdi.gov.ro/news-dezbatere-publica-proiecte-de-cercetare>
- "Challenges - Change" ["Provocări - Schimbare" - PPS]: <https://uefiscdi.gov.ro/news-dezbatere-publica-pachet-informatii-provocari-schimbare-pps>

These requirements and criteria will be included in other funding instruments – where appropriate.

- **Support for universities aimed at developing institutional capacity for the implementation of open science practices, through the Institutional Development Fund (FDI).** Included in the specific *Domain 6: Development of institutional capacity for research in higher education institutions; OG 6.6. Support for activities carried out by higher education institutions for the development of university research.* FDI aims to support institutional development projects at public universities in Romania, funded from the Ministry of Education's budget, with the purpose of increasing institutional capacity and improving the quality of the national higher education system.  
More information: <https://www.cnfis.ro/finantare/dezvoltare-institutionala/>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

## 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy<sup>iii</sup>, strategy or plan of action on science, technology and innovation (STI)?

**YES** / NO

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

### Title of the Policy:

RO version: *Strategia Națională de Cercetare, Inovare și Specializare Inteligentă 2022-2027* (<https://www.mcid.gov.ro/programe-nationale/strategia-nationala-de-cercetare-inovare-si-specializare-inteligenta-2022-2027>)

EN version: *National Strategy for Research, Innovation, and Smart Specialisation (SNCISI) 2022-2027* (<https://uefiscdi.gov.ro/resource-867892-SNCISlenglish-version.pdf%20>)

**Responsible:** Ministry of Research, Innovation and Digitalisation (MCID) (former); as of 2025, the Ministry of Education and Research and the National Research Authority (under the subordination of the Ministry of Education and Research)

**Authority in Charge of Development:** UEFISCDI for the *Ministry of Research, Innovation and Digitalisation (MCID)*, in the context of the European Structural and Investment Funds (ESIF) project: *"Increasing the capacity of the RDI system to respond to global challenges"* (SIPOCA 592 | MySMIS Code 127557)

**Stakeholders Involved in the Development Process:** Regional Development Agencies, universities and research institutes, private sector stakeholders

Year of Adoption: 2022; year of Last Update: 2022

### Key Areas/Sections of the Policy:

The National Strategy for Research, Innovation, and Smart Specialisation (SNCISI) 2022-2027 aims to enhance Romania's research and innovation ecosystem, ensuring alignment with European and global trends. Implemented through the **National Plan for Research, Development, and Innovation (PNCDI IV)**, it provides structured financing to support RDI activities.

### Key Objectives:

1. **Strengthen RDI System:** Enhance research capacity, infrastructure, and human resources.

2. **Support Innovation Ecosystems:** Foster collaboration between research organizations and the private sector, focusing on smart specialisation areas.
3. **Mobilize Towards Innovation:** Encourage public-private partnerships and technology transfer to drive economic growth.
4. **Enhance International Collaboration:** Integrate Romania into European and global research networks.

**Thematic Priorities:**

- Digitalization, Industry, and Space – AI, robotics, and space technologies.
- Climate, Energy, and Mobility – Renewable energy and sustainable transport.
- Food, Bioeconomy, and Natural Resources – Sustainable agriculture and biodiversity.
- Health and Well-being – Medical innovation and personalized healthcare.
- Culture, Creativity, and Inclusive Society – Social resilience and cultural heritage.
- Civil Security and Resilience – Cybersecurity and disaster risk reduction.

This strategy reinforces **open science principles**, and introduces a *specific objective dedicated to Open Science: Objective 1.2. Ensuring the transition to open science and facilitating the road towards excellence in scientific research*. The actions in this objective are:

- Promoting and ensuring open access to the publicly funded research publications (ensuring the eligibility of the necessary costs);
- introducing DMPs requirements (ensuring FAIR data and open access to research data should be promoted according to the principle as open as possible as closed as necessary and the costs associated with the research data management will be considered eligible costs; the research infrastructures should receive support for the development and implementation of DMPs associated with experiments);
  - Of course, there may be legitimate reasons for delayed or restricted access to certain data, such as those related to, for example, national security, personal data, confidential data, data protected by intellectual property rights, trade secrets but there should be a balanced approach on open access to research data.
- promoting and supporting the existing and new infrastructures, digital repositories and services that supports open science to ensure the long-term storage of open access publications and research data and onboarding into disciplinary databases and EOSC;
- citizen science - supporting projects that encourage citizen involvement in various stages of the research process, such as data collection;
- establishing and implementing a national support mechanism for the transition to open science;
- developing of open science specific skills and competences of researchers and personnel of academic and research institutions;
- rethinking of the research assessment system so it considers also open science practices (adoption of new research and career evaluation metrics).

**White Book on Open Science Transition in Romania (2023-2030)**

**Title of the Policy:**

RO version: *Cartea Albă a Tranziției către Știința Deschisă 2023-2030* (<https://www.open-science.ro/resurse/cartea-alba-a-tranzitiei-catre-stiinta-deschisa-2023-2030>)

EN version: *The White Book of the Transition to Open Science (2023-2030)* (<https://www.open-science.ro/resurse/the-white-book-of-the-transition-to-open-science>)

**Authority in Charge of Development:** UEFISCDI for the *Ministry of Research, Innovation and Digitalisation (MCID)*, in the context of the European Structural and Investment Funds (ESIF) project: *“Increasing the capacity of the RDI system to respond to global challenges”* (SIPOCA 592 | MySMIS Code 127557)

**Stakeholders involved in the Development Process:** Researchers and RDI community stakeholders; UNESCO Chair on Science and Innovation Policies – SNSPA; the Romanian Open Science Cloud Initiative (RO-NOSCI);

Year of Adoption: 2022; Year of Last Update: 2023

**Key Areas/Sections of the Policy:**

The White Paper was developed in correlation with international provisions and developments on

OS (among these, the UNESCO Recommendation on Open Science 2021, the EC Recommendation (EU) 2018/790 of 25 April 2018 on access to and preservation of scientific information, the Council conclusions on Research assessment and implementation of Open Science of June 2022).

The White Paper on the Transition to Open Science (2023-2030) presents the vision, the strategic objectives and necessary actions for the implementation of open science vision in Romania, in alignment with international and European open science policies and recommendations. The development of the White Paper is based on a consultation process initiated by UEFISCDI in 2020 and organized in several stages, through a series of workshops dedicated to identifying Romania's strategic options on open science and dissemination events organized within the NI4OS, OpenAire and other projects, as well as 2 national surveys developed by the Executive Agency in 2021 on the understanding of the needs and familiarity with the open science practices among the academic and research community.

8 Strategic Objectives have been included in the White Paper on the Transition to Open Science (2023-2030), with related actions proposed to be implemented at national level:

1. Ensuring open access to scientific publications resulting from publicly funded research;
2. Research data management and ensuring open access to research data;
3. Ensuring transparency and fairness of open access publishing costs and those of access to international publications;
4. Developing the infrastructure and services for open science;
5. Establish a long-term governance for the transition to open science and EOSC;
6. Developing the skills needed to implement open science;
7. Adapting the process of research assessment and rewarding in the new context of open science;
8. Citizen science.

**Link to the Webpage:** <https://www.open-science.ro/resurse/cartea-alba-a-tranzitiei-catre-stiinta-deschisa-2023-2030>

- does it include any of the following open science elements:

- open access to scientific knowledge **YES**
- open science infrastructure **YES**
- open engagement of societal actors **YES**
- open dialogue with other knowledge systems **YES**

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)<sup>iv</sup> at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h*)

YES / **PARTLY** / NO

**If yes or partly**, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

The **White Paper aligns well with several UNESCO Open Science principles**, particularly regarding **open access, public funding mandates, and governance discussions**. However, it lacks explicit coverage of **multilingualism and indigenous knowledge protection**.

*a. Ensuring that open science incorporates values and principles to ensure shared and reciprocal benefits, avoiding unfair extraction of data and knowledge*

The white paper emphasizes **equity in access to scientific information**, addressing issues related to the fairness of open science practices. It also references **the need to prevent unintended negative consequences of open science**, such as high publication costs leading to inequalities

*b. Ensuring publicly funded research follows open science principles and its outputs are openly licensed or dedicated to the public domain*

The document strongly supports **open access to publicly funded research** and **FAIR (Findable, Accessible, Interoperable, Reusable) data principles**. It states that access to research publications and data should be provided **as early as possible and free of charge**, in alignment with European and international recommendations

*c. Encouraging bibliodiversity, including diverse formats, business models, and humanities/social sciences*

The document acknowledges the importance of a diversity of open access publishing models (e.g., Diamond Open Access), and mentions the need to encourage bibliodiversity.

*d. - Encouraging multilingualism in scientific publications and academic communications*

The document does not significantly address multilingualism in open science. Its focus is primarily on accessibility, data openness, and publishing frameworks, but there is no dedicated section on promoting research in multiple languages.

*- Protecting the rights of indigenous communities, including traditional knowledge, per the 2007 UN Declaration*

There is no direct mention of indigenous rights, traditional knowledge, or the role of indigenous communities in open science practices.

*h. Enabling open multi-stakeholder discussions on open science challenges (competition, exploitation, IPR, privacy, security, inequalities)*

The strategic document discusses the challenges of open science related to intellectual property, data security, highlights the importance of stakeholder discussions and the role of governance bodies in overseeing open science implementation

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

2.3 In your country, are there specific policy instruments<sup>v</sup> that aim to promote open science?

**YES**/ UNDER DEVELOPMENT/NO

**If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

**The National Plan for RDI 2022-2027 (PNCDI IV), the main instrument for implementing the [The National Research, Innovation and Smart Specialisation Strategy \(SNCISI\) 2022-2027](#)**

Types of support/ projects proposed to be funded within the time horizon 2022-2027:

Open science is one of the ten principles mentioned in the plan and the National Plan includes RDI financing programs and types of projects proposed to be financed and which support open science:

- developing the institutional capacity to implement open science practices (grants for the development of OS skills),
- open access publication (APCs)
- preparation/ management of the research data (RDM),

- supporting the participation of RO and the relevant RO entities in/ for the development of European Open Science Cloud - EOSC;
- exchange of best practices for the development of capabilities needed to implement open science (grants);
- Science and Society Programme:
  - o Projects for the training of researchers in science communication;
  - o Science in schools,
  - o Festival of science, creativity and innovation,
  - o Science and technology summer schools,
  - o Supporting national and international scientific events.
  - o Innovation in citizen science (supporting exploratory activities for the development of specific innovative methods and methodologies in order to engage citizens in science, awarding successful projects with citizen science components)

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

YES / UNDER DEVELOPMENT / **NO**

**If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

**As recent changes occurred in the governmental structures, decisions regarding this action will be possible after the UNESCO survey closes.**

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

YES / **UNDER DEVELOPMENT** / NO

**If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#) included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

**Partial monitoring of institutional policies** for Open Science has indeed been carried out by **UEFISCDI** through various consultations, in different projects and mainly in associated to the development of the National Strategic Framework for the Transition to Open Science (in 2021 - 2023). For example, at the end of 2023, a survey among RO-NOSCI members identified that two institutions (one university and one research institute) had developed an institutional Open Science policy and six others (four universities and 2 research institutes) declared having the policy under development. The information has not been updated as of now.

The National Research, Innovation and Smart Specialisation Strategy (SNCISI) 2022-2027 (OS.1.2.) includes the development of a Governance and Monitoring mechanism for Open Science which is not in place yet.

- 2.6 In your country, are there specific funding mechanisms<sup>vi</sup> for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

**YES** / UNDER DEVELOPMENT / NO

**If yes or under development**, please provide details and links as relevant.

APCs are eligible costs in publicly funded research: According to the Gov. Decision no. 134 of February 16, 2011, which approves the Methodological Norms for determining categories of expenses for research and development activities and innovation stimulation financed from the state budget, the publishing costs are considered eligible expenses.

RDMS costs are eligible costs in the National Research, Development and Innovation Plan (PNCDI IV).

“Stiinta in scoli” funding instrument (PNCDI IV) to be launched in near future. It enhances STEAM education through hands-on learning, team competitions, and science debates, lab visits, expert lectures, and Fab-Lab student projects. More details: <https://uefiscdi.gov.ro/stiinta-in-scoli-ss-sc>.

The Anelis Plus consortium in Romania (an association comprising Romanian universities, research and development institutes, and central university libraries) supports the Open Access publishing through transformative agreements with major publishers. These agreements enable researchers from member institutions to publish their work openly without additional fees, as the publication costs are covered by the consortium. The list of publishers with whom agreements have been established includes Elsevier, IEEE, Royal Society of Chemistry (RSC), Institute of Physics (IOP), Wiley, Cambridge University Press, Springer, and Sage. More information on the Anelis Plus website: <https://anelis-plus.ro/open-access/acorduri-transformative-2024/>

Apart from these, the actions mentioned already at 2.3. planned/ proposed in The National Plan for RDI 2022-2027 (PNCDI IV).

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

- 2.7 In your country, have open science practices<sup>vii</sup> been integrated in existing research funding mechanisms?

**YES** /NO

**If yes**, please provide details and links as relevant.

- In order to encourage the uptake of Open Science practices and to put into practice the mandatory provision of the National Strategy, regarding the elaboration of data management plans (DMPs), compliant with FAIR principles within the RDI projects financed with public funds, OS related elements have been included as requirements in the funding instrument Centres of Excellence (<https://uefiscdi.gov.ro/centre-de-excelenta-coe-coe2023-747>). In the project description applicants are asked to describe OS practices used in the project and correspondingly, evaluators will assess applications, considering also the description provided for OS practices. These provisions regarding DMPs and FAIR data were for the first time requested within any RDI funding programme.
- The requirements and criteria have been extended to other funding instruments which are planned to be launched in near future:
  - Complex Frontier Research Projects: <https://uefiscdi.gov.ro/news-dezbatere-publica-proiecte-de-cercetare>
  - "Challenges - Change" ["Provocări - Schimbare" - PPS]: <https://uefiscdi.gov.ro/news-dezbatere-publica-pachet-informatii-provocari-schimbare-pps>

These requirements and criteria will be included in other funding instruments – where appropriate.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.i, 23.c)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

The monitoring framework for the SNSCI strategy is detailed in the Chapter Monitoring (pages 105-109 - the English version, [link](#)).

Establishing and implementing a national governance and monitoring support mechanism for the transition to open science is envisaged in the SNCISI (under the Objective OS.1.2.) as follows:

Under the supervision of an Open Science Council which should be created as part of the Ministry of Research (currently the National Research Authority, under the subordination of the Ministry of Education and Research), this mechanism will provide support for:

- the definition and implementation of open science-specific policies, the
- coordination of the development and implementation of the
- recommendations of the Strategic Document regarding the framework for
- the development and of open science in Romania;
- the development of specific open science capabilities;
- the coordination of the expert network at the level of research organisations;
- the coordination of the development of the management capacity for open
- science at the level of research organizations

Objective 5 from the *White Paper on the Transition to Open Science* for the time horizon 2023-2030 details the core provisions of the strategy, further expanding on the actions needed to ensure a coherent Open Science Governance and a robust monitoring mechanism while facilitating the national transition to open science.

**The National Research, Innovation and Smart Specialisation Strategy (SNCISI) 2022-2027 provides** indicators from the SNCISI Nomenclature associated with the national target (ANNEX 1 of the strategy, for English version - [link](#)), and corresponding to the **OS.1.2. Ensure the transition to open science and facilitate the consolidation of excellence in scientific research:**

- S14 – Share of Web of Science (WoS) publications in Open Access
- IS-P1 – Number of Web of Science (WoS) indexed articles, including those published in Open Access and international scientific co-publications
- IS-P2 – Number of scientific articles with research datasets published in Open Access

### 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

According to the latest available data, Romania's research and development (R&D) expenditure was 0.52% of its Gross Domestic Product (GDP) in 2023. This information was published by Eurostat in December 2024 and pertains to the year 2023.

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

As of early 2024, Romania's internet penetration rate stood at 91.6%, with approximately 18.06 million internet users. This data was published by DataReportal in their "Digital 2024: Romania" report.

In terms of household access, 88.6% of Romanian households had home internet access in 2024, reflecting an increase of 2.9 percentage points from the previous year. This information was reported by the National Institute of Statistics (INS) in December 2024. Regarding internet bandwidth, Romania boasts high-speed connectivity. The average fixed download speed in the country reached 257 Mbps, surpassing countries like Germany (113 Mbps) and Austria (107 Mbps). This data was highlighted in a study reported by Romania-Insider.com.

- 3.3 Are there national research and education networks (NRENs)<sup>viii</sup> active in the country? (ref.: [\(iii\) 18.c](#))

**YES** / NO

**If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

RoEduNet, managed by the Agency of Administration of the National Network for Education and Research (AARNIEC). RoEduNet provides data communication services to major academic institutions, research centers, and cultural institutions across the country. RoEduNet is a member of the European research and education network GÉANT, facilitating high-speed connectivity and collaboration with other NRENs across Europe.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

- 3.4 Are there national or institutional [open science infrastructures](#)<sup>ix</sup> in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

**YES** / UNDER DEVELOPMENT / NO

**OS national portal:** [www.open-science.ro](http://www.open-science.ro) (non-commercial)

**Doctoral thesis** (thesis starting 2016, 19.663 available as of February 2025): <https://rei.gov.ro/teze-doctorat>. (non-commercial)

**BrainMap:** <https://www.brainmap.ro>. It serves as an online community for researchers, innovators, technicians, and entrepreneurs, aiming to connect expertise within Romania, fostering collaboration and sharing of research outputs. The platform supports dissemination and capacity-building actions related to open science practices at the national level. (non-commercial)

**EERTIS** (European Research and Technology Infrastructures System) <https://eertis.eu>: an online platform that unites public and private research and technology infrastructures. It serves as a single-entry point, offering open access to a wealth of information about these infrastructures and their associated services and equipment. EERTIS aims to bridge the gap between service providers and potential users, fostering collaboration and innovation across various fields. (non-commercial)

#### **Digital repositories (non-commercial)**

- **RODA - Romanian Social Data Archive**

**Description:** The national institution in Romania specializing in archiving electronic data collections from social research.

**Repository Link:** <http://www.roda.ro/en/home>

- **Erdélyi Digitális Adattár (EDA)**

**Description:** Established by the Transylvanian Museum Association to develop and maintain collections and promote education and research. Interfaces available in Romanian, Hungarian, English, and German.

**Repository Link:** <https://eda.eme.ro>

- **ECOLIB Digital Repositor**

**Description:** The institutional repository of the National Research and Development Institute for Industrial Ecology (ECOIND), providing access to articles, theses, conference papers, and project descriptions.

**Repository Link:** <http://dspace.incdecoind.ro>

- **Digital Repository of the "Lucian Blaga" Central University Library of Cluj-Napoca**

**Description:** Contains books, manuscripts, periodicals, and iconographic materials from the library's collections. Searchable in metadata and full text, with interfaces in Romanian, English, German, and Hungarian.

**Repository Link:** <http://dspace.bcuccluj.ro/>

- **ASPECKT Dspace**

**Description:** The institutional repository of Transilvania University of Braşov, hosting intellectual work produced by academic and research staff, including theses and, in the future, technical reports and conference papers.

**Repository Link:** <http://aspeckt.unitbv.ro/jspui>

#### **Citizen Science Platforms (non-commercial):**

- The Citizen Science platform developed by Babeş-Bolyai University, promotes citizen involvement in various stages of the research process, including data collection. Through this online platform, the university aims to present its range of projects that engage citizens as comprehensively and transparently as possible. <https://citizenscience.openscience.ubbcluj.ro/home>
- *Citizen Science Vest* is also an innovative platform developed by the West University of Timişoara, designed to engage science enthusiasts and curious individuals across Western Romania. <https://citizenscience.uvt.ro/en>

#### **Other National initiatives that support the EOSC integration and Open Science:**

- **CLOUDIFIN – Cloud Computing Infrastructure at IFIN-HH**

Cloud infrastructure managed by the **Department of Computational Physics and Information Technology (DFCTI)** at IFIN-HH. It provides **high-performance computing (HPC)** and **high-throughput computing (HTC)** resources, supporting large-scale international collaborations such as **WLCG, EOSC, and ELI-NP**. More info: <https://cc.ifin.ro/cloudifin/index.html>

- **CLOUDUT – Research Cloud Infrastructure in Technical University Cluj-Napoca**

Cloud infrastructure to support research in areas like big data and deep learning. This initiative aligns with Open Science principles by providing accessible computing resources

for the research community.

More info: <https://cloudut.utcluj.ro/en/>

- **UBB SmartCloudDSS – Decision Support System at Babeş-Bolyai University**

A cloud-based decision support system providing accessible, data-driven tools for researchers. This initiative fosters data sharing and collaborative analysis,.

More info: <https://cercetare.ubbcluj.ro/ro/dezvoltarea-infrastructurii-cloud-a-universitatii-babes-bolyai-cluj-napoca-pentru-realizarea-unui-sistem-integrat-de-management-academic-si-suport-decizional-bazat-pe-bigsmart-data-smartclouddss/>

- **MOISE – Modernized Computing and Storage Infrastructure at West University of Timişoara**

Upgraded infrastructure for cloud and high-performance computing (HPC) services, enabling complex analyses and data-sharing.

More info: <https://www.uvt.ro/>

- **CYRUS – Medical Cloud Infrastructure at University of Medicine and Pharmacy "Iuliu Haţieganu"**

An integrated **private and public cloud infrastructure** providing Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS) for medical research. It includes a Big Data application, improved access for universities and research institutions worldwide through public cloud solutions, enhanced data security and disaster recovery, and reduced costs for acquiring essential research applications.

More info: <https://www.cloudumfcluj.ro/>

- **High-Performance Cloud Platform at Politehnica University of Timişoara**

A cloud-based infrastructure supporting advanced research activities through scalable computing resource.

More info: [https://upt.ro/Informatii-utile\\_high-performance-cloud-platform-at-upt\\_556\\_en.html](https://upt.ro/Informatii-utile_high-performance-cloud-platform-at-upt_556_en.html)

- **Cloud and Big Data Infrastructure at University of Craiova**

- Enhances data processing capabilities and facilitating researchers' access to international scientific research resources through the infrastructure created and based on the cooperation agreements signed between university and partner institutions

More info: [https://www.ucv.ro/pdf/media/stiri/2020/06/Comunicat-de-presa-incepere-proiect\\_2.pdf](https://www.ucv.ro/pdf/media/stiri/2020/06/Comunicat-de-presa-incepere-proiect_2.pdf)

- **Cloud and Big Data Infrastructure at Maritime University of Constanţa**

A project integrating cloud computing and big data solutions for maritime research, encouraging shared knowledge, **integrated into international cloud structures and large-scale data infrastructures**, fostering global collaboration and advanced research.

More info: <https://cmu-edu.eu/blog/2023/05/12/comunicat-de-presa-proiect-cloud/>

**Romanian National Open Science Cloud Initiative (RO-NOSCI):** Established in June 2021, RO-NOSCI is a consortium dedicated to offering support for developing the infrastructure for data sharing and collaboration among Romanian research institutions. The initiative was initiated by UEFISCDI, the [National Institute for Research and Development in Informatics](#) – ICI Bucharest and [The National Research and Development Institute for Physics and Nuclear Engineering "Horia Hulubei"](#) - (IFIN-HH).

The strategic Objective 1.2. „Ensuring the transition to open science and facilitating the road towards excellence in scientific research” of the National Strategy on Research (SNCISI) promotes and supports the existing and the new initiatives for open science (such as infrastructure, digital repositories, services), in particular for the long-term storage of research data and scientific publications, for the integration into disciplinary databases and/or in EOSC.

**The Research Infrastructure funding Program (not yet launched) of PNCDI IV 2022-2027** envisages support for the participation of Romania and Romanian entities in the development of the **European Open Science Cloud (EOSC)**. The aim is the development and integration of infrastructures, services and digital repositories for open science in the EOSC, in accordance with the policies of the European Commission.

**If yes or under development**, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society – Citizen Science Platforms, institutional digital repositories
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

**For each infrastructure**, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? ([ref.: \(iii\) 18.e, 18.f](#))

YES / **NO**

**If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science<sup>i</sup>, stage of research, accessibility to all the relevant users in the region or internationally).

**If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / **NO**

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? ([ref.: \(iii\) 18.g](#))

YES / **NO**

**If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science<sup>i</sup>, stage of research, accessibility to all the relevant users in the region or internationally).

#### 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? ([ref.: \(iv\) 19.a, 19.c](#))

**YES** / NO

If **yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

- UEFISCDI, as a partner, in the context of the [FAIR-IMPACT project](https://www.fair-impact.eu/fair-impact-open-calls-support), fosters knowledge exchange and offers structured support, expert guidance for the development of FAIR implementation plans at national level. The programme targeted organisations with a national scope, included (but not limited to) those mandated by EOSC. More information: <https://www.fair-impact.eu/fair-impact-open-calls-support>
- Several training sessions have been organized part of the different projects where the experts in the OSKH are involved, such as:
  - (Online) „**FAIR national plan in Germany and national initiatives to support FAIR in Romania**”  
Organized by FAIR-Impact Project  
Date: February 1, 2024  
Number of Attendees: around 20  
Link: [Event Details](#)
  - Training Session "**Open Science – Policies and Practices**" organized by the Regional Center for Research Career Guidance and Counseling – Bucharest - Ilfov region, focusing on open science policies and practices, emphasizing the importance of sharing scientific results with society.  
Date: December 2nd, 2024  
Number of Attendees: around 20  
Link: [Event Details](#)
  - (Online) "**Building Consistency for Open Science in Europe – Rhetorics and Practices**"  
Organized by: UEFISCDI as the host of the Open Science Knowledge Hub and the [Romanian Programme Operator for the Research Programme under EEA and Norway Grants](#), with the support of the UNESCO Chair on Science and Innovation Policies – SNSPA, Bucharest and Scientific Knowledge Services  
Date: April 2021  
Number of Attendees: around 500, people from 36 countries  
Link: [Event Details](#)
  - (Online) **Training Session on Open Science, EOSC, FAIR Data** (part of the NI4OS project)  
Organized by UEFISCDI & ICI-Bucharest  
Date: January 28, 2021  
Link: [Event Details](#)
- **Courses** on Open Science for PHDs candidates, organized through the UNESCO Chair on Science and Innovation Policies (Open Science Lab) at SNSPA (December 2022, December 2023, December 2024)
- Mentorship on Open Science related topics to develop policy recommendations  
Organized by the UNESCO Chair within the CNFIS-FDI-2023-F-0565 project, this program aimed to develop public policy accelerators in Romania. It was supported by the Student Entrepreneurial Society of SNSPA and recommended by UEFISCDI's Innovation Café. More information: <https://snsa.ro/au-inceput-inscrierile-pentru-programul-policy-accelerator-hub-2023>
- Foreseen :  
Part of a recently started project at national level, financed through the European Social Fund (ESF) and national funds: [Adapting the educational offer to labor market demands through evidence-based policy making](#). Within the project, a specific activity is carried dedicated to **co-creation of recommendations for a competency framework needed**

**for open science, involving hands-on training sessions.** The Open Science Lab at the UNESCO Chair on Science and Innovation Policies will be involved and also universities from different regions of the country.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES / **NO**

**If yes**, please provide more information and links as relevant.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

**Recommendations for capacity building programmes for different categories of stakeholders will be developed within the project** [Adapting the educational offer to labor market demands through evidence-based policy making](#), Activity 4.4. **Research on the development of emerging skills, competencies, and professional profiles necessary for open science**; financed through the European Social Fund (ESF) and national funds. <https://uefiscdi.gov.ro/adaptarea-ofertei-educationale-la-solicitarile-pietei-muncii-prin-fundamentarea-politicilor-din-domeniu-pe-baza-de-date>

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: [\(iv\) 19.b](#))

**YES** / NO

**If yes**, please provide more information and links as relevant.

Foreseen in a recently started project at national level, financed through the **European Social Fund (ESF) and national funds**: *Adapting the educational offer to labor market demands through evidence-based policy making*. Within the project, a specific activity is carried dedicated to **Research on the development of emerging skills, competencies, and professional profiles necessary for open science (A4.4, 18 months, starting February 2025)**. The main outcomes:

- 1) **Analysis of policies and practices** (both nationally and internationally) regarding the development of skills and competencies necessary for open science: **managing FAIR scientific information**, open access to research results, the use of specific standards and protocols, management of open educational resources, etc.;
- 2) Analysis of the **current state of skills and existing educational and training offerings at the national level** relevant for training specialists—such as IT experts in data-intensive computing, technicians, engineers, and data administrators (data stewards, data curators, etc.);
- 3) **Co-creation of recommendations for a competency framework needed for open science**: building the skills & capacities for students, researchers, academic and research institution staff, and other relevant actors in the private and public sectors, as well as civil society, to implement open science

More information: <https://uefiscdi.gov.ro/adaptarea-ofertei-educationale-la-solicitarile-pietei-muncii-prin-fundamentarea-politicilor-din-domeniu-pe-baza-de-date>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 4.4 Are open educational resources<sup>x</sup> as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: [\(iv\) 19.d](#))

YES/ PARTLY/ **NO**

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: [\(iv\) 19.e](#))

**YES** / NO

If yes, please provide more information and links as relevant.

Romania promotes science communication through **public and private initiatives**, fostering scientific literacy, interdisciplinary collaboration, and public engagement.

#### **Public Initiatives**

##### **1. "Știința în Școli" (Science in Schools)**

Led by UEFISCDI under PNCDI IV, this program enhances **STEAM education** through:

- Hands-on learning, team competitions, and science debates,
- Lab visits, expert lectures, and Fab-Lab student projects.

More details: <https://uefiscdi.gov.ro/stiinta-in-scoli-ss-sc>

##### **2. "Research in the Spotlight" Podcast – UEFISCDI & EEA & Norway Grants**

A podcast featuring **scientists discussing innovations in biotech, environment, and medicine**.

- **Recognized by Science Europe** as a best practice in science communication.
- Covered by major media:

- Market Watch:

[http://www.marketwatch.ro/articol/18084/Research\\_in\\_the\\_spotlight\\_-\\_un\\_instrument\\_util\\_pentru\\_comunicarea\\_stiintei/](http://www.marketwatch.ro/articol/18084/Research_in_the_spotlight_-_un_instrument_util_pentru_comunicarea_stiintei/)

- Agerpres: <https://www.agerpres.ro/ots/2024/02/07/science-europe-a-selectat-un-podcast-creat-in-romania-ca-exemplu-de-buna-practica-in-comunicarea-stiintei--652942>

More details: <https://uefiscdi.gov.ro/research-in-the-spotlight>

#### **II. Private Initiatives**

##### **1. Science Communication Newsletter – Mădălina Cocea**

Provides insights into scientific discoveries and communication strategies.

More details: <https://www.comunicarestiintifica.ro/despre/>

##### **2. Summer School on Science and Technology**

Offers hands-on research experiences for students and educators.

More details: <https://msciteh.educatiepentrustiinta.ro/>

##### **3. "Game of Science" – Science Communication Competition**

Trains young scientists in public communication, ending in a presentation competition.

More details: <https://www.gamesofscience.org/>

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

## 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i\)](#)

YES / NO

If yes, please provide more information and links as relevant.

The dialogue about rethinking the existing research assessment and evaluation processes has already started at national level since the development of the *White Book of the Transition to Open Science*. During the consultation on the *Green Book of the Transition to Open Science* (18 August-30 September 2022) out of 186 respondents, 88% indicated that Recommendation 7 (final form as the Objective 7 in the White Paper - *Adapting the Research Evaluation and Rewarding Process to the New Context of Open Science*) is highly or very highly relevant to their work, 8% stated that this recommendation has medium relevance, and 4% specified that its relevance is low or very low. The respondents emphasize the need for fundamental changes in the evaluation system, with clear and realistic criteria prioritizing public dissemination of research.

A key element of this transformation is international collaboration. Many believed that integrating foreign experts into the evaluation process would enhance credibility and align research standards with those of the European Union. At the same time, transparency must become a cornerstone of the system. Removing confidentiality from the evaluation process—both for reviewers and authors—was seen as a crucial step toward accountability, helping to eliminate conflicts of interest and ensuring fairer assessments.

Yet, beyond evaluation reforms, respondents stress the pressing need for increased research funding and a revised system for rewarding researchers. Open science practices should not be implemented in isolation but should be part of a broader shift in how research is valued and incentivized.

Despite this optimism, concerns linger. Some warned that changing evaluation and reward systems alone will not be enough - these shifts must be accompanied by systematic reforms. More discussions on these topics should be encouraged, as they are crucial for ensuring meaningful progress. Others proposed a practical approach: opening funding lines for pilot programs to test new evaluation methods before implementing sweeping changes (as it started in 2023 with the funding instrument “Centers of Excellence” described at the Q 1.4.). However, there is also a note of caution - some research areas deal with sensitive information, and in such cases, national interests must be safeguarded.

Not all respondents were in favor of the recommendations related to research evaluation in the Green Book; major obstacles like inadequate funding and a shortage of qualified evaluators were mentioned. They also express deep concerns over intellectual property rights and the risks of prematurely sharing sensitive research data. Without proper safeguards, they argue such openness could do more harm than good.

As the debate continues it is clear that the path to reform is complex. While many see open science as the future, its success depends on careful implementation, sufficient resources, and a balanced approach that considers both progress and protection.

The report of the consultation on the Green Book of the transition to Open Science: <https://www.open-science.ro/resurse/raport-al-rezultatelor-consultarii-privind-cartea-verde-a-tranzitiei-catre-stiinta-deschisa-2022-2030>

The National Strategy promotes rethinking research assessment and the *Objective 7 of the White Paper on the Transition to Open Science* is dedicated to describing actions to start *Adapting the process of research assessment and rewarding in the new context of open science*. These can be summarized as follows:

**Reviewing Evaluation Processes:** Assess current research evaluation methods (for researchers, projects, and institutions) to determine how well they support and encourage open science practices.

**Developing New Evaluation Criteria:** Update or create new evaluation methods, tools, and indicators in collaboration with the research community to recognize and reward open science practices, including early data sharing, open collaboration, public access, and broader research outputs.

**Ongoing Monitoring and Review:** Periodically assess evaluation processes to ensure effectiveness and efficiency.

**Transparency and Communication:** Ensure clear and accessible information about evaluation methods, tools, and criteria, with active involvement from relevant stakeholders.

**Support Actions - Working Groups & Events:** Establish working groups and organize events to facilitate dialogue and consultation on evaluation reforms.

**Support Actions - Guidance Materials:** Develop informational guides to help research organizations update their evaluation systems and implement recognition and reward mechanisms for open science practices.

In order to continue the dialogue towards rethinking research assessment, discussions are organized with the national community, such as the **Rethinking Research Assessment Conference**, organized on November 14, 2024 by UEFISCDI, in collaboration with the Open and Universal Science project (OPUS), Science Europe, the Coalition for Advancing Research Assessment (CoARA), and the UNESCO Chair on Science and Innovation Policies (SNSPA). [Event Details](#).

Coalition for Advancing Research Assessment (CoARA): 25 institutions from Romania are members of the CoARA

Also a CoARA member, UEFISCDI has recently developed its [Action Plan for CoARA for the period between 2025 and 2028](#). The document outlines the Agency's commitment and (planned) actions meant to contribute to the alignment of its research assessment practices, as well as of the ones from national level, with the principles of the Agreement on Reforming Research Assessment (ARRA). It focuses on implementing activities around a series of main elements, such as narrative CVs, the responsible use of research metrics, recognizing Open Science and Research Data Management practices in funding calls, a new format for research outputs list and a limited number for outputs, and other.

So far, evaluation criteria related to Data Management Plans (DMPs) and Open Science have been included in the funding instrument "Centers of Excellence". Applicants have been required to describe how open science practices (e.g. Open Access to publications, research data management, citizen science, and other) are implemented in their project to enhance its objectives. If no open science practices are applied, a justification is required. Additionally, the applicants have been asked to describe the research data management (RDM) aligned with FAIR principles (Findable, Accessible, Interoperable, and Reusable). If funded, a Data Management Plan (DMP) is required within the first six months of the project's implementation. More info: <https://uefiscdi.gov.ro/centre-de-excelenta-coe-coe2023-747>

These requirements and criteria have been extended to other funding instruments which are planned to be launched in near future:

- Complex Frontier Research Projects: <https://uefiscdi.gov.ro/news-dezbatare-publica-proiecte-de-cercetare>
- "Challenges - Change" ["Provocări - Schimbare" - PPS]: <https://uefiscdi.gov.ro/news-dezbatare-publica-pachet-informatii-provocari-schimbare-pps>

These requirements and criteria will be included in other funding instruments – where appropriate.

In addition, until 2027, UEFISCDI is also implementing activities that aim to reflect on existing good practices for performance based research funding mechanisms addressed to higher education institutions, and subsequently provide recommendations for such potential mechanisms that could be of interest at national level. In order to do so, it will look

at research assessment practices in a coherent way, trying to consider the alignment of practices for all three types of evaluands – researchers, project proposals and institutions. The activity will involve a large consultation process at national level with research communities and Open Science practices will be discussed together with a variety of types of other contributions to research as potential contributions that can be recognized in PBRF mechanisms. This initiative is carried part of the project *Romania on the horizon 2030: Increasing international relevance through the internationalization of higher education* (Activity 4), funded through ESF and national funds: <https://uefiscdi.gov.ro/romania-la-orient-2030-cresterea-relevantei-internationale-prin-internationalizarea-invatamanului-superior>

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

**YES** / NO

**If yes**, please provide details and links as relevant.

It's a partial yes, as currently, career evaluation and progression systems in Romanian academia acknowledge especially Open Access but with significant restrictions, primarily emphasizing quality over accessibility. These are based on the recently adopted national minimum standards for the conferment of the title of associate professor, full professor, and habilitation certification were adopted through a ministerial order published on Tuesday, February 12, in the *Official Gazette of Romania*. (Official source: [Monitorul Oficial al României](#)).

While Open Access publishing is supported, journals that charge mandatory publication fees are often excluded from evaluation criteria (e.g., Mathematics, Informatics). The Chemistry Commission allows fee-based journals but limits their proportion within an applicant's publication record.

Beyond Open Access, other Open Science practice (such as open data, open peer review, and citizen science) - are not explicitly incorporated into career assessments. The only notable exception is in Mathematics, where preprints on arXiv.org are recognized as a legitimate way to share research. However, collaboration with non-scientific actors, interdisciplinary knowledge-sharing, and engagement beyond academia are not included in evaluation frameworks, nor are there specific provisions to introduce them by 2025. The primary focus across scientific commissions remains on scientific impact, citations, and adherence to ethical research standards, rather than broader Open Science principles.

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

**YES** / NO

**If yes**, for each initiative, please provide more information, including on the stakeholders<sup>xi</sup> that are involved or lead the initiative, and links as relevant.

Open science practises are recognized in the evaluation phase of certain research grants:

- The Centers of Excellence: <https://uefiscdi.gov.ro/centre-de-excelenta-coe-coe2023-747>
- Complex Frontier Research Projects: <https://uefiscdi.gov.ro/news-dezbatere-publica-proiecte-de-cercetare>
- "Challenges - Change" ["Provocări - Schimbare" - PPS]: <https://uefiscdi.gov.ro/news-dezbatere-publica-pachet-informatii-provocari-schimbare-pps>

**If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 5.4 Have there been any unintended negative consequences<sup>xii</sup> of open science practices reported in your country? (ref.: [\(v\) 20](#))

**YES** / NO

**If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

In 2021, debates have sparked around the rising trend of publishing in Open Access in MDPI. The public debates were carried about the quality and impact of such publications, especially considering MDPI's rapid expansion and the proliferation of some special issues, which some argue may compromise academic standards. Additionally, discussions on certain media and opinions platforms have delved into the broader implications of open access publishing and concerns have been raised about the financial burden placed on institutions and the potential shift in focus from research quality to quantity, driven by the need to cover publication fees and achieve higher output metrics. These developments underscore the need for a balanced approach to open access publishing in Romania, ensuring that the drive for accessibility does not compromise research integrity or lead to unsustainable financial practices.

## 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

**YES**, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

**If yes**, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects

- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.

- promoting open innovation practices.

- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

## 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a\)](#)

- The national efforts in the transition to Open Science are deeply connected with global efforts, strengthening international collaborations, mainly through the centralized support offered by UEFISCDI, through the Open Science Knowledge Hub. Open Science Knowledge Hub (OSKH) at UEFISCDI is the OpenAIRE NOAD, RDA Node and the main contact point for OS in Romania, implementing practices in support of OS mainly part of Horizon Europe and national funded projects.

Key initiatives and collaborations:

- **Engagement in EU relevant Projects (past and current)** supporting FAIR data and open science practices :
  - GraspOS (HORIZON-INFRA-2022-EOSC-01-01)
  - OPUS (HORIZON-WIDERA-2021-ERA-01-45)
  - SECURE- Sustainable Careers for Research Empowerment (HORIZON-WIDERA-2021-ERA-01);
  - FAIR-IMPACT (INFRA-2021-EOSC-01-05)
  - NI4OS-Europe,
  - OpenAIRE Advance
- Collaboration with organisations and initiatives driving the Open Science transition :
  - OpenAIRE (as OpenAIRE NOAD Romania),
  - RDA (as RDA Node Romania);
  - Member of the EOSC Association;
  - Member in CoNOSC (aligning open science policies in Europe), and representation in the CoNOSC Board
  - Member of the Science Europe Working Groups on Open Science and Research Culture.
  - Member of CoARA
  - DORA signatory.
  - Endorser of the Action Plan for Diamond Open Access
  - Part of the UNESCO Working Group on Open Science, close collaboration with the UNESCO Chair on Science and Innovation Policies – SNSPA, Bucharest.
  - Representation in the IAU Working Group on Open Science

More details:

[UEFISCDI Open Science Hub](#)

<https://unesco-chair-snsipa.com/>

[www.open-science.ro](http://www.open-science.ro)

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b, 22.e, 22.f](#))

YES / **NO**

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: [\(vii\) 22.c](#))

YES / **NO**

If **yes**, please provide links as relevant and more information.

If **no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

## 8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/**NO**

If **yes**, please provide more information.

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- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.  
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.  
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

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- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))
- xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.
- xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.